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Project Title:	Response of Thyroid Functional Tissue Units to Space-Like Radiation Fields		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
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Human Research Program Risks:	None		
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Comments:	PI is deceased as of January 2010.		
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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The thyroid like most epithelial-based glandular tissues are a common site of tumor development and sensitive to ionizing radiation induced carcinogenic changes. Using thyroid as a model, especially the well-defined normal cell line that we have characterized, provides us the opportunity to investigate how normal tissue responds to ionizing radiation from different sources, doses and dose rates, at multiple levels. From “Radiation Hazards to Crews of Interplanetary Missions” a major question raised was “Are there studies that can be conducted to increase the confidence of extrapolation from rodents to humans of radiation-induced genetic alterations that in turn could enhance similar extrapolations for cancer?” We have established tissue analogs that we believe will increase the certainty of this type of extrapolation. The thyroid tissue analog produced during growth in a low-shear bioreactor is a realistic three-dimensional replica of the tissue being modeled. The bioreactor system provides stable environmental controls not achievable by other methods. Thus, our goal is to confirm and extend the characteristic pattern of radiation modulated

Task Description:	gene expression in rat thyroid tissue analogs using space-like low dose/low dose rate radiation, and challenge the chronically irradiated tissue analogs. To- Confirm and extend the characterization of the pattern of radiation modulated gene expression in rat thyroid tissue analogs using space-like low-dose/low-dose rate radiation of low, medium and high linear energy transfer (LET). We hypothesized that in the LEO analog a low-dose/low-dose rate “priming” exposure to radiation would invoke a suite of cell and tissue level homeostatic mechanisms that would be reflected in gene expression profiles and complementary structural and functional changes in component cells, as well as in their functional tissue units (FTU’s, i.e. follicles). Once the thyrocytes have completed the “transition” to the “primed” state, the thyroid FTU’s will have an altered response to an acute challenge exposure to radiation that will take the form of increased resistance to biological changes relative to “unprimed” cells. The response of the FTU’s to the GCR exposures is less predictable as no information comparable to our system is available. However, based on our previous work we expect that principal regulatory elements in the priming transition will include transforming growth factor-beta and protein kinase C isoforms. These important products will direct other respondent genes that will be identified by microarray expression profiling. It is expected that the transcription factors, oncogenes and cell coordinating gene products will exhibit the most significant alterations. The tissue analogs will be subjected to various dose and delivery schedules, of gamma rays and protons (LLU), as well as carbon and iron particles (NSRL, BNL), to obtain gene expression profiles that reflect critical changes in structure and functional aspects of thyroid homeostasis, or normalcy. Generally these studies address the following -- Thyroid tolerance, radiation dose limit to head & neck, susceptibility of cancer in normal tissue. Additionally these studies address several points indicated in the “Strategic Proposed Plan for Space Radiation Health Research” (1998). What mechanisms are involved in modulating radiation damage at the molecular level (repair enzymes, etc.)? What is the probability of initiating neoplastic cell transformation or other steps leading to a cancerous cell? How do cellular repair mechanisms modulate damage produced by ionizing radiation? And how can the radiation effects on cells in culture be related to radiation effects in “normal cells”?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information gained by studying radiation effects will be translatable to risk assessment for assigning limits to radiation workers, pre-dosing for more effective radiotherapy and the consequences of long duration space flight. Additionally, and more globally, better understanding of the biological response to radiation will improve our ability to treat individuals exposed to radiation as a consequence of accidents, terrorism, radiotherapy, and their occupation.
Task Progress:	This task progress section will be elaborated using the three newly numbered critical questions that this research addresses. Critical question 30a - what are the probabilities for degenerative tissue risks from protons and HZE ions as a function of NASA’s operational parameters (age at exposure, age and time after exposure, gender, tissue, mission, radiation quality, dose rate)? Our research addresses tissue, radiation quality and to a lesser degree the dose rate components of this question. Our studies indicate that protons and HZE particles (carbon and iron) have unique signatures and result in more dramatic responses to low dose radiation than gamma rays. In fact the data results showed that protons have the greater response, in terms of the number of modulated genes, than carbon or iron ions. In the studies of acute responses and those of the challenge exposures indicated that protons behave more like HZE particles than they do gamma-rays or low LET radiation. In our previous studies using flask grown cultures we found that in most aspects of radiobiologic endpoints protons were more damaging than gamma-rays, even at higher energies of protons that yield a LET very similar to that of ?-rays the differences were dramatic (Green et al., 2001 a; 2001b; 2002a; 2002b and 2005). The findings of several of our experiments reveal protons and carbon were more affecting of gene expression than gamma-rays or iron ions. Suggesting that they have the potential to cause greater effects than would be predicted based on their intermediate LET’s/fluence. Critical question 30b -what are the mechanisms of degenerative tissues risks in the endocrine system? The level of analysis conducted with gene arrays has the potential to direct studies leading to a better mechanistic understanding of the response to low dose radiation, although these studies do not confirm mechanistic function. We have spun-off a number of findings (protein kinases C, tissue inhibitor of metalloproteinase-3, c-fos, etc.) from the gene expression data into other projects that are being conducted by four gradate students. The results of 15,900 gene microarrays are very labor intensive, but once conducted the data base allows for re-analysis to address new questions that arise to investigate particular gene or sequences that are found to participate in newly emerging pathways. Thus, continued investigation with the data we generated will lead to a better understanding of mechanistic questions. Critical question 30e -what quantitative procedures of theoretical models are needed to extrapolate molecular, cellular, or animal results to predict degenerative tissue risks in astronauts? This is the main critical question addressed by the research conducted in this grant. A significant proportion of historic radiobiological data has been collected from transformed cell lines that may not accurately represent normal cell/tissue responses. Currently there is a paradigm shift toward utilization of normal tissue models in radiobiologic investigations as well as toward reducing doses and dose rates (Brooks 2005, Nelson & Bissell, 2005). The goal is to complete as many of the categories as possible in the diagram below and solve for the ultimate objective (in vivo human) Rats- In Vitro -->Tissue Analogs -->Primary Explants -->Animals (in vivo) Humans- In Vitro -->Tissue Analogs -->Primary Explants -->Humans (in vivo) Our three-dimensional tissue analog model represents an intermediate state between the isolated follicles found in flask-grown cultures and the highly organized multifollicular assembly of the intact thyroid gland. The identification of differences in gene expression found when comparing FRTL-5 cells grown in flasks with those grown in bioreactors supports the biological differences measured previously using these culture conditions (Green et al., 2002b). The results indicate that the tissue analogs and flask-grown follicles have substantially dissimilar responses to radiation stress. Thus, the method of culturing (flasks vs. bioreactors) dramatically impacts how these cells respond to ionizing radiation. It is likely that cells in the bioreactor cultures and the flask cultures represent different microenvironments that are controlled by different regulatory factors, and in turn respond differently to ionizing radiation. From this perspective, it will still be necessary to determine which of these gene expression patterns represent the best approximation of the radiation response in vivo. However, the previous reports of normal tissue-like properties of engineered organoids suggest that our bioreactor-generated thyroid tissue will exhibit radiation-induced changes in gene expression that more closely

	approximates the in vivo condition. Thereby providing a reasonable approximation to in vivo and thus reducing the need for animals in ascertaining risk. In conclusion, the results support that at low doses of radiation the gene expression response of tissues is significantly less than would be predicted from cells grown in vitro. Therefore, risk based on in vitro data will lead to an overestimation of the risk of exposure to low dose (< 1Gy) radiation. Secondly, the responses to relatively high doses resulted in gene expression patterns that were not significantly different whether the cells were grown in flasks or as tissue analogs. This suggests that extrapolation of data from high dose; high dose rate experiments are less dependent on higher order cellular integration than lower dose responses. Our laboratory is currently further investigating the contributions of higher order tissue organization to radioresponsiveness and improved ability to extrapolate information from in vitro tissues to in vivo models.
Bibliography Type:	Description: (Last Updated: 01/29/2015)
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